

Vortex-Cooled Rocket Engine

Vortex Thermal Insulation Quantification Under Variable Oxidizer Mass Flow

Department of Mechanical and Aerospace Engineering
University of California San Diego

Team 14

Anthony Gaeta, Gregorio Zaltzman, Sainiray Mantrala

June 12, 2026

Indices

Abstract.....3

Introduction.....3

Methodology.....4

Results.....7

Discussion.....16

References.....18

Appendices.....18

Abstract

Vortex-cooled rocket engines use oxidizer injection to generate a vortex and swirling flow field that forms a protective layer between the chamber wall and the hotter flow near the core. This cooling system offers a potential simple and low cost alternative to regenerative cooling, making it extremely appealing for propulsion systems within student-led rocketry initiatives. This study investigates the thermal dissipation effectiveness of the Project Maelstrom vortex cooled rocket engine developed by the UC San Diego Rocket Propulsion Laboratory. Computational Fluid Dynamics (CFD) simulations with heat transfer analysis were performed to simulate the flow field and heat transfer through the chamber walls. A study was conducted through three conditions, maximum, half, and quarter flow conditions to evaluate the influence of oxidizer mass flow rate on vortex formation and cooling effects. Velocity fields, streamline patterns, and temperature distributions were examined to quantify the effectiveness of oxidizer vortex cooling. The results showed the formation of a stable vortex and indicates that the vortex provided measurable thermal cooling by maintaining lower temperature near the chamber walls. These findings support the feasibility of vortex cooling as a passive thermal management technique and will provide quantitative support for the Maelstrom team's cold-flow test.

Introduction

The use of computational grids in numerical methods are essential to high-fidelity computation fluid dynamics. Fine, well-balanced meshes are critical when simulating real-world aerospace hardware, particularly when the systems operate at extreme thermal and pressure conditions. Ideally, all meshes could be made infinitely small to produce accurate results, but computation expense scale inversely with decreasing cell size and thus striking a careful equilibrium between simulation realism and computational constraints presents a paramount engineering fluid mechanics problem.

Vortex-cooled rocket engines are a class of rocket propulsion systems in which the combustion chamber wall is thermally protected from the hot exhaust not through conventional active cooling circuits but rather through the fluid mechanics of a co-axial vortex structure. In this novel design, the four oxidizer inlets are tangentially injected into the base of the combustion chamber through peripheral inlets, which generates a swirling vortex of cold oxidizer traveling axially toward the nozzle. This annular flow acts as a thermal barrier that separates the host combustion products in the core of the structure from the outer wall. This system offers smaller scale, student-built propulsion systems, an attractive cooling system with lower complexity and cost compared to regenerative cooling systems.

Project Maelstrom is a senior design capstone project conducted in collaboration with UC San Diego's student-led Rocket Propulsion Laboratory (RPL). The project's goal is to evaluate the overall feasibility of the vortex-cooled engine design for use in student-built rocketry. The engine's main structure is manufactured from acrylic, a material that softens and loses structural integrity at sustained temperatures of 370 K thus making thermal management especially important. This study's purpose is to support the Maelstrom's upcoming experimental cold-flow test by providing expected thermal and kinematic data to compare experimental measurements against.

This work specifically conducts a parametric CFD investigation of the vortex insulation effectiveness as a function of oxidizer inlet mass flow rate. Beyond just focusing on grid convergence of the velocity field, this study aims to quantitatively test the cooling capabilities of the vortex system across flow conditions. This study evaluates the RANS solver with Conjugate Heat Transfer through the acrylic wall and systematically varies the oxidizer-to-fuel mass flow ratio across three select flow settings: maximum flow, half flow, and quarter flow. By measuring the area-weighted average temperature of the chamber wall and the chamber fluid temperature across these conditions, the vortex shielding's efficiency can be determined across decreasing mass flow. Obtaining quantitative performance data of the maelstrom vortex rocket engine's cooling efficiency across different mass flow envelopes is critical to ensuring the team can run the cold-flow test at the optimal operating envelope.

Methodology

Geometry and CAD preparation

To prepare the model for simulation, the CAD geometry was imported into ANSYS SpaceClaim where the fluid domain was extracted using the Volume Extract Tool. The created fluid domain spanned the interior of the thrust chamber through the converging-diverging nozzle and into the exit plenum. The fuel inlet injectors extra lines and surfaces were simplified using SpaceClaim's Simplify tool as the injector inlet area is not the quantity of interest in this investigation. Named group selections were also created in SpaceClaim prior to meshing to ensure correct automatic boundary condition assignment in Fluent. These groups included: *fuel_inlets*, *oxidizer_inlets*, *nozzle_outlet*, and *outer_wall*.

Numerical Solver

This experiment makes use of the Ansys 2025 Suite with a combination of SpaceClaim, Fluent Meshing, and Fluent Solution. The meshes and simulations were performed through UC San Diego's EBU2 Lab desktops which make use of more processing power

thus enabling for time independent simulations. The density-based solver was selected over pressure-based alternatives due to the compressible nature of the flow at the throat and exit which exceeded 0.3 thus making density variations non-negligible and pressure–velocity coupling assumptions of the pressure-based solver insufficient. The Roe-FDS flux model was employed for its handling of contact discontinuities and strong pressure gradients within compressible and highly turbulent flow. Second-order upwind discretization was used for the momentum, turbulent kinetic energy, turbulent dissipation rate, and energy equations, while gradients were computed using the Least Squares Cell-Based method.

Physical Models

Flow physics assumptions were established based upon known physical operating conditions. The working fluid was modeled as a compressible gas, with density computed from the ideal gas law. Both specific heat and thermal conductivity were held at constants of 0.0242 W/m K and 1006.43 K/kg K respectively. Viscosity was modeled using Sutherland’s three coefficient law rather than constant value due to the compressibility of the flow and its superior performance in flows with large temperature gradients. Due to the Navier-Stokes Equations having no analytical solution for turbulent regimes, the k-epsilon numerical model was employed to approximate the true value of turbulence. The k-epsilon model’s computational efficiency is ideal for this simulation, coupled with its high accuracy deriving from the two additional transport equations: one for the turbulent kinetic energy, and one for its rate of dissipation [1]. This allows the effects of turbulence on the mean flow to be captured in a highly accurate manner without requiring high-scale processing power linked with resolving all scales of motion.

The energy equation was enabled to resolve the thermal field and Conjugate Heat Transfer was activated by assigning the acrylic wall the thermal properties listed in Table 1. This allowed the solver to compute both fluid convection and solid conduction through the chamber wall simultaneously. Initial conditions were established upon design specification of the previously-designed engine. These include the mass flow rate through the fuel and oxidizer inlets, though the oxidizer mass flow rate was adjusted as per the investigation’s goal. Additionally, the maximum pressure allowed at the inlets was determined by machinery restrictions and a maximum design mach number greater than 2.54 was set for the nozzle. [2]

Table 1: Material properties used in the simulation for the fluid and solid domains

Fuel		Oxidizer		Nozzle	
Mass flow rate [kgs ⁻¹]	0.02376	Max Mass flow rate [kgs ⁻¹]	0.04752	Ma_{Nozzle}	> 2.54
Pressure [kPa]	250	Pressure [kPa]	350	Pressure [kPa]	N/A
Temperature (K)	293.3	Temperature (K)	293.3	Temperature (K)	N/A
Viscosity (Ns/m)	Sutherland	Viscosity (Ns/m)	Sutherland	Viscosity (Ns/m)	N/A

Mesh preparation and geometry

In order to both conduct and validate this experiment, a grid convergence study was performed, keeping parameters such as operating conditions, boundary conditions and turbulence methods the same while changing mesh parameters such as local sizing, surface mesh sizing, max cell length and boundary layers. Adjusting these mesh parameters enabled the creation of three meshes with three different grid resolutions whose parameters can be seen in Table 2 below. From each mesh: the number of cells, maximum skewness, minimum orthogonality and mesh run time were saved and recorded in Table 3 as well as a corresponding cross-plane image of each mesh visible in Figure 1. After running each simulation for each given mesh, the following parameters were saved: *Chamber Plane Area-Average Temperature, Cross Plane Area-Average Temperature, Nozzle-Plane Area-Average Temperature, Plenum Plane Area-Average Temperature, Solid Acrylic Wall Area-Average Temperature, Solid Acrylic Wall Facet Max Temperature, and Area-Average Fluid Facet Max Velocity.*

Table 2: Preliminary mesh sizing parameters across all three studied meshes.

	Level 1	Level 2	Level 3
Local Sizing (mm)			
Oxidizer Inlet	-	-	0.1
Fuel Inlet	-	-	0.1
Surface Mesh (mm)			

Minimum Size	0.4010	0.2005	0.1
Maximum Size	10.2	5.1330	2
Volume Mesh - Polyhedra (mm)			
Boundary Layers	2	3	6
Max Cell Length	15.0	6.6227	3

Results

Table 3. Mesh Generation Results output from Ansys Fluent Meshing across Level 1 (Coarse), Level 2 (Medium), and Level 3 (Fine) meshes.

Mesh Refinement Level	Cell Count	Maximum Skewness	Min Orthogonality	Generation Time
Level 1 (Coarse)	327,970	0.763	0.107	0.32 minutes
Level 2 (Medium)	590,087	0.773	0.200	0.46 minutes
Level 3 (Fine)	1,644,765	0.770	0.140	0.95 minutes

Grid Convergence Study

A three-level mesh refinement study was performed to evaluate the sensitivity of the numerical solution to grid resolution. The coarse, medium, and fine meshes contained 327,970, 590,087, and 1,644,765 cells respectively. As shown in Table 3, all meshes maintained acceptable quality metrics, with maximum skewness values below 0.8 and minimum orthogonality values remaining above 0.1. Mesh generation times increased with refinement level, but remained computationally manageable for all cases.

The peak velocity across all meshes did not vary by more than 0.01 m/s or 0.0015%. This near identical convergence across all three mesh levels confirms that the kinematic solution for this simulation is grid independent. For the average wall temperature, across all three mesh sizes the temperature did not vary by more than 0.005 K also indicating that the convergence across all mesh sizes has become fully grid independent. This result is particularly important because it confirms the CHT coupling where the wall temperature converges simultaneously with the velocity field facilitated by the inflation layer that successfully resolves the boundary layer.

Table 4: Grid convergence study results where Facet Maximum Velocity of the Fluid is measured from the Ansys Fluent simulation for all three mesh levels.

Mesh Level	Max Velocity (m/s)	Average Acrylic Wall Temperature (K)
Level 1 (Coarse)	666.900	296.263
Level 2 (Medium)	666.900	296.263
Level 3 (Fine)	666.912	296.268

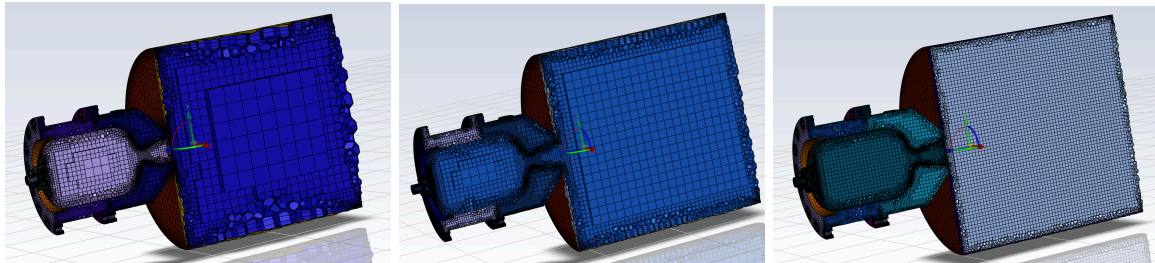


Figure 1: Meshes of the Project-Maelstrom vortex-cooled rocket engine for the coarse level 1 mesh (left), medium level 2 mesh (middle), and fine level 3 mesh (right).

Velocity Field Analysis

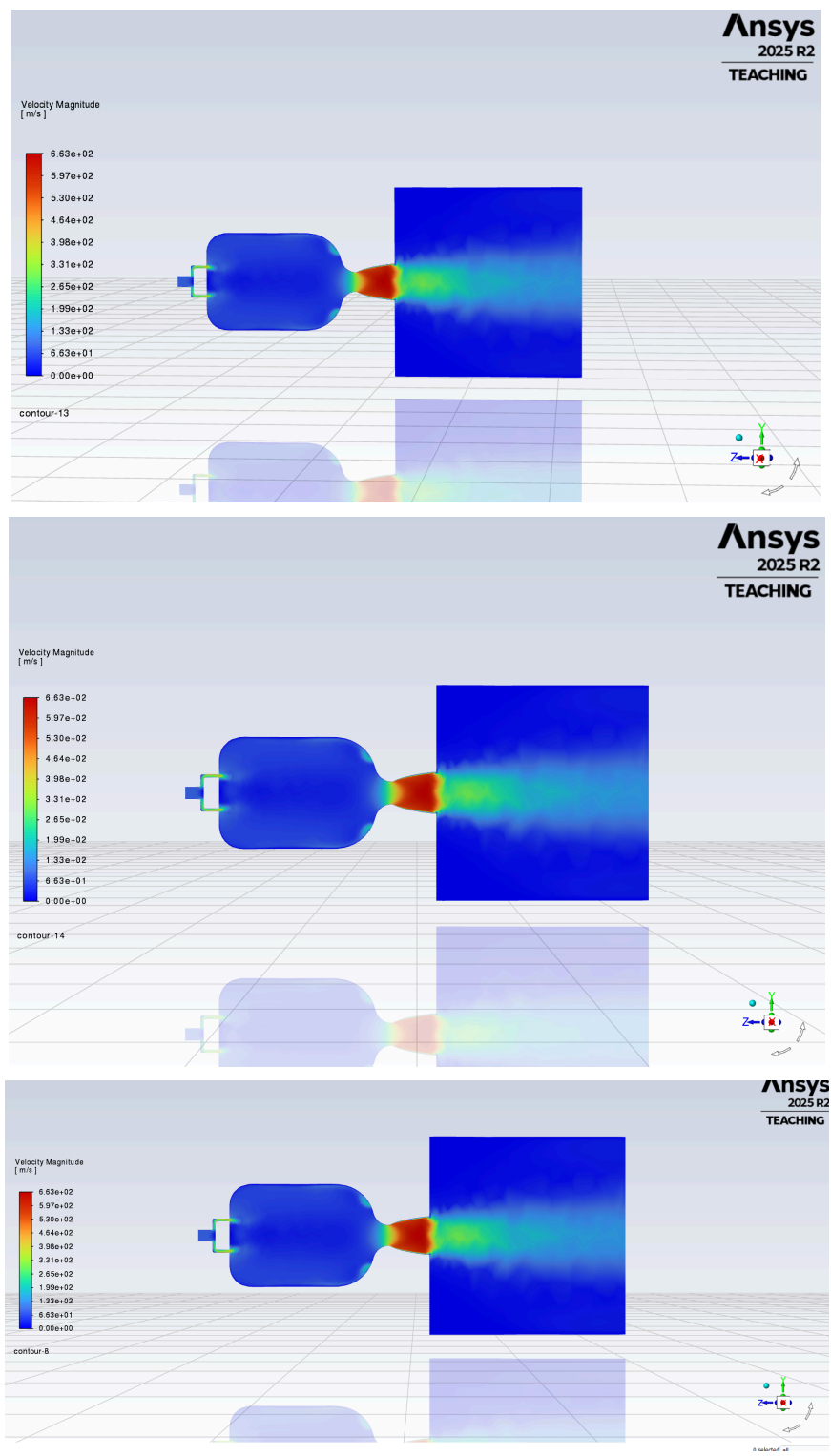


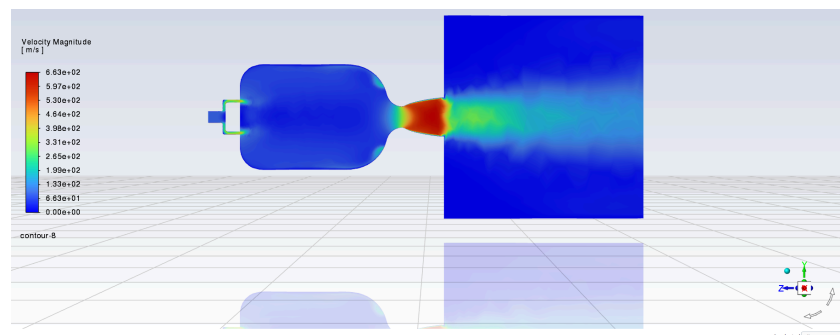
Figure 2: Velocity (max flow) magnitude contour of the Project-Maelstrom vortex-cooled rocket engine for the coarse level 1 mesh (top), medium level 2 mesh (middle), and fine level 3 mesh (bottom). The velocity is in units of meters per seconds, with blue being the smallest value of velocity and red being the largest.

Visually, the velocity contours showed little differences between the medium and fine mesh solutions. The overall flow structure, nozzle acceleration, and chamber vortex max velocity values also remained very close to each other across the meshes. This indicated that the solution is largely independent from the mesh resolution. Therefore, the Level 3 mesh was used for analysis in the following sections, though simulations at different flow conditions were run for all 3 meshes and can be found in **Appendix B**.

Velocity magnitude contours for the three mesh levels are shown in Figure 2. In all cases, flow enters the chamber through the fuel and oxidizer inlets before accelerating through the converging-diverging nozzle. The highest velocities occur near the nozzle throat and exit, where the flow experiences rapid acceleration due to the reduction in cross-sectional area.

The similarity of the velocity contours across all mesh levels demonstrates that the primary flow features are captured even on the coarser grids. This further supports the conclusion that the numerical solution has reached grid independence.

Effect of Oxidizer Mass Flow Rate on Velocity Distribution



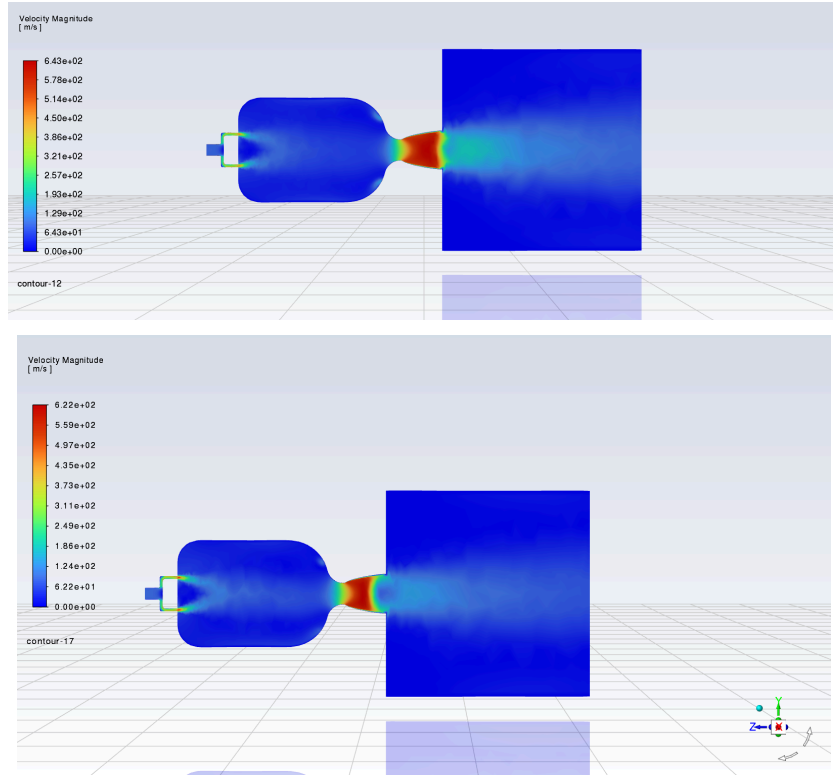


Figure 3: Velocity magnitude contour of the Project-Maelstrom vortex-cooled(fine mesh) rocket engine for full full flow (top), half flow (middle), and quarter flow (bottom)

To study the impact of oxidizer mass flow rate on vortex cooling, simulations were performed at maximum, half, and quarter oxidizer flow conditions using the fine mesh. For all three operating conditions, the flow accelerated through the nozzle and expanded into the plenum region. However, one noticeable feature was that reduced oxidizer mass flow led to weaker chamber flow speed.

Vortex Formation and Pathline Analysis

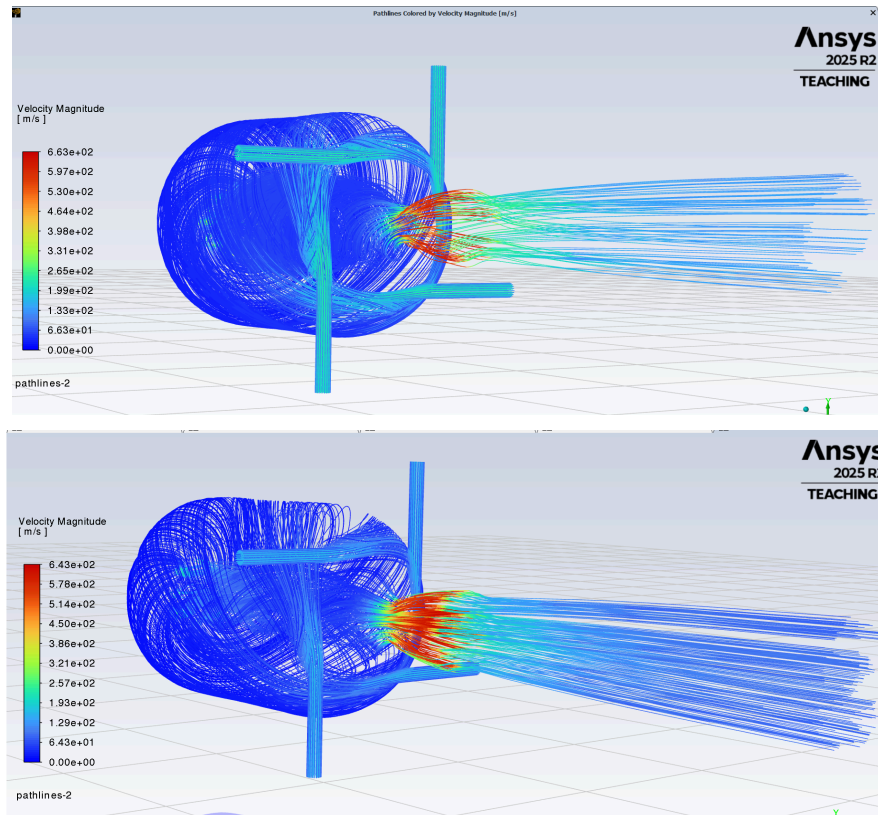


Figure 4: Velocity Pathlines for max flow (top) half flow (bottom)

Velocity pathlines are presented in Figure 4 for the full-flow and half-flow operating conditions. The pathlines clearly show the formation of a swirling flow structure generated by the tangential oxidizer injection ports. After entering the chamber, the oxidizer follows a spiral trajectory around the chamber walls before moving toward the nozzle.

This observed behavior is consistent with the guiding principle of the vortex-cooled engine, with oxidizer pathlines that remain concentrated near the outer chamber walls. This spatial separation creates the potential for a thermal barrier between the chamber wall and the hotter core region.

*Thermal Performance and Chamber Cooling***Table 5:** (Area-weighted average temperatures measured at several locations throughout the engine)

Measureme nt Location	Maximum Flow Rate	Half Flow Rate	Quarter Flow Rate
Chamber Plane (Area-Avg)	315.29 K	320.10 K	322.51 K
Cross Plane (Area-Avg)	302.13 K	304.84 K	313.12 K
Nozzle Plane (Area-Avg)	147.39 K	149.50 K	224.09 K
Plenum Plane (Area-Avg)	299.03 K	300.09 K	304.57 K
Solid Acrylic Wall (Area-Avg)	297.27 K	300.72 K	307.85 K
Solid Acrylic Wall (Facet Max)	399.97 K	399.97 K	399.97 K

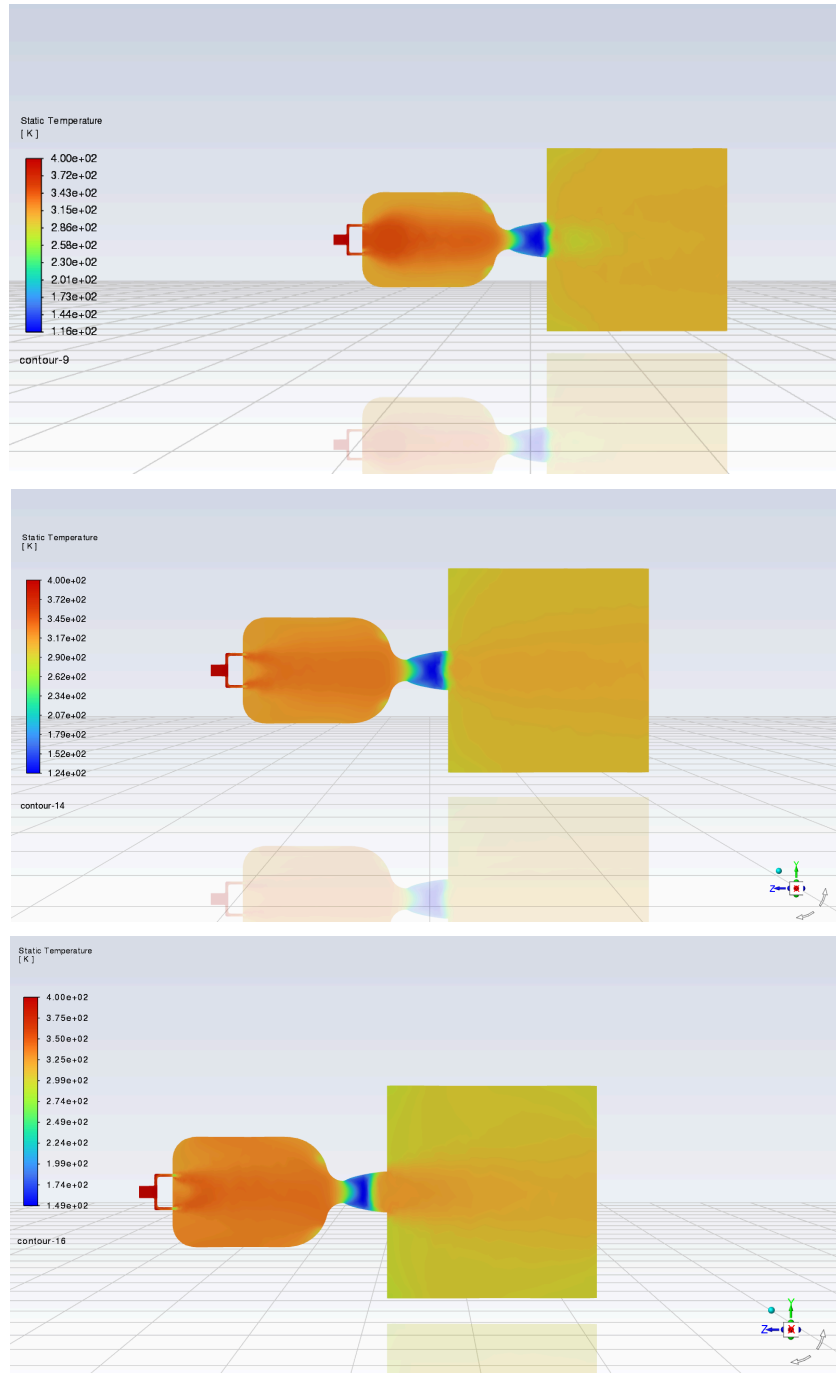


Figure 5: Temperature magnitude contour of the Project-Maelstrom vortex-cooled(finest mesh) rocket engine for full flow (top), half flow (middle), and quarter flow (bottom).

The results show a clear relationship between the thermal properties of the engine and oxidizer mass flow rate.

The chamber-plane temperature increased from 315.29 K at maximum flow to 320.10 K at half flow and 322.51 K at quarter flow. Similar increases were observed at the cross-plane and

plenum-plane measurement locations. These trends indicate that lower oxidizer mass flow weakens the vortex structure, reducing the ability to maintain a cool near-wall layer that thermally isolates the chamber wall from the hotter core flow.

The most important thermal metric for this study is the average temperature of the acrylic wall. Under maximum-flow conditions, the wall temperature remained at 297.27 K. When oxidizer flow was reduced to half flow, the average wall temperature increased to 300.72 K. Under quarter-flow conditions, the average wall temperature increased further to 307.85 K.

The difference between the maximum-flow and quarter-flow wall temperatures was 10.58 K, demonstrating that stronger oxidizer injection provides measurable thermal protection to the chamber wall. These results directly support the hypothesis that the vortex layer acts as an insulating barrier that reduces heat transfer from the chamber core to the surrounding structure.

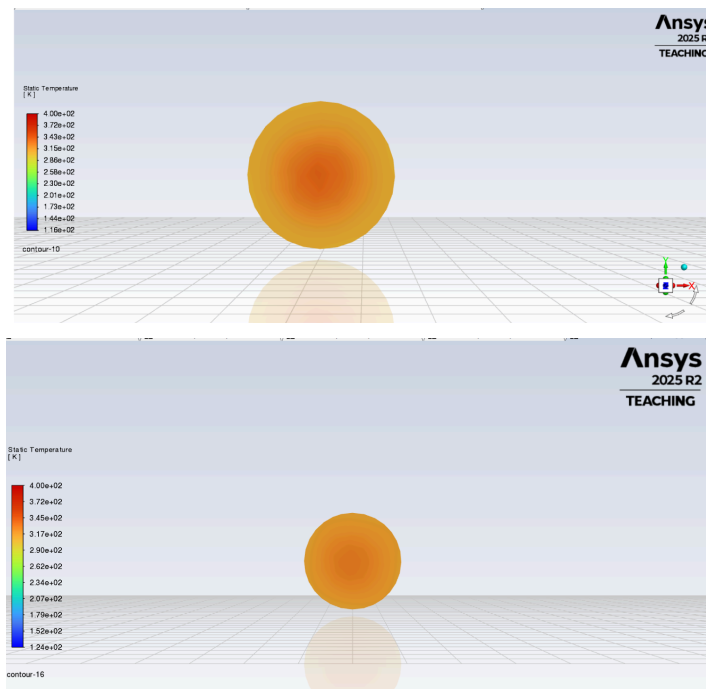


Figure 6: Temperature magnitude contour of the plane located in the chamber of Project-Maelstrom rocket engine (finest mesh). Full flow (top), and half flow (bottom).

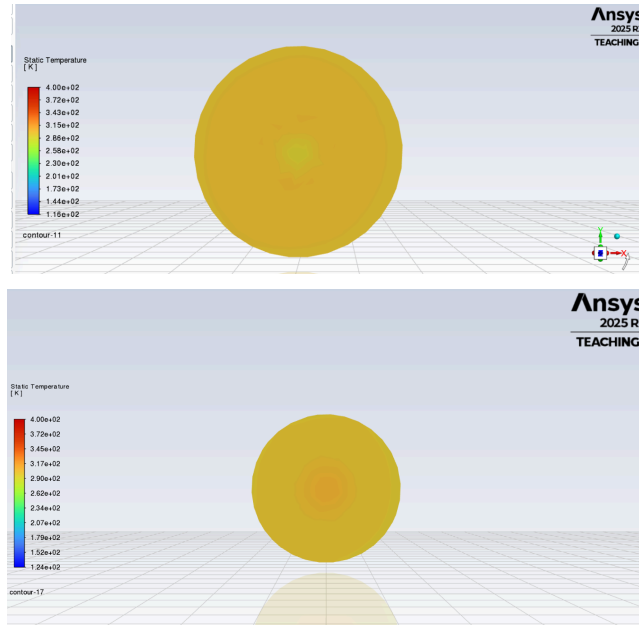


Figure 7: Temperature magnitude contour of the plane located in the plenum of Project-Maelstrom rocket engine (finest mesh). Full flow (top), and half flow (bottom).

Temperature contours for the two of the three operating conditions are shown in Figures 6 through 7. The contours reveal a unique temperature distribution within the chamber. Higher temperatures are concentrated near the center of the chamber, while lower temperatures are maintained near the wall region.

This radial temperature gradient is consistent with the expected behavior of vortex cooling. The tangentially injected oxidizer remains near the outer chamber walls and forms a cool layer surrounding the hotter chamber core. As oxidizer mass flow decreases, this barrier layer becomes less effective and chamber wall cooling is reduced.

Temperature contours taken on chamber cross-sections further illustrate this effect, with a larger temperature difference between the chamber center and the wall region under maximum oxidizer flow conditions.

Overall, both the quantitative temperature measurements and the contour visualizations demonstrate that the oxidizer vortex provides measurable thermal insulation. Increasing oxidizer mass flow strengthens the vortex structure and improves the engine's ability to maintain lower wall temperatures.

Discussion & Conclusion

Thermal Field and Vortex Insulation Quantification

The primary objective of this study is to quantify the vortex-induced thermal insulation with respect to changing oxidizer mass flow rate. With the fuel inlet being at 400 K and the oxidizer inlets being at 300 K, the temperature gradient in the chamber represents the stark difference in thermal energy between the hot core and the cooler annular vortex. The solid wall's area-weighted average temperature is able to be measured through the CHT formulation.

On the finest mesh and at maximum flow, the average wall temperature was measured to be 297.27 K as can be seen in **Table 5**. Compared to the 400 K temperature of the core, this result clearly demonstrates that under designed max flow conditions, the tangential oxidizer vortex is highly effective at insulating the acrylic wall despite the 100K temperature gradient between the core and the wall.

Reducing the oxidizer mass flow to half, raised the fine-mesh average wall temperature to 300.72 K, indicating a 3K increase compared to max-flow conditions. This is a central finding of this study, as it demonstrates that under sub-optimal oxidizer flow rate conditions produces a measurable increase in average wall temperature. This makes sense as lower mass flow rates means the incoming cool oxidizer has less momentum, which allows more of the central hot core to diffuse radially outward and increase contact with the chamber-wall boundary layer.

Reducing the oxidizer mass flow to one quarter of the intended max-flow conditions, raised the average wall temperature to a significant 307.85 K. At this temperature, the acrylic wall can expect to lose its structural integrity and the rocket engine can no longer be expected to function at its designed operating level. It is evident that there is a non-linear relationship between the oxidizer mass-flow-rate and the thermal cooling capabilities of the vortex with an initial 3K jump for a 50% reduction followed by a 7K jump in temperature when a further 50% mass-flow rate reduction is applied. The design implications for the Maelstrom project are clear, the engine should not be operated at oxidizer flow rates below 50% as once below that level, the acrylic can expect to receive a sustained temperature of over 300K.

There are several limitations to the current setup that need to be considered. The injector plate geometry was simplified to a uniform inlet to remove the influence of mixing on the near-inlet field. The simulation was run using a steady-state RANS solver, which time-averages any steady vortex elements that may be actually present in the flow. Finally, the 400K fuel inlet temperature is a simplified model for simulating thermal combustion characteristics and is a significant underestimate of real

combustion temperatures. Actual flame temperatures can produce much stronger temperature gradients along the chamber walls, making the current model a preliminary understanding of vortex cooling rather than an applicable example.

Future work can work to address these limitations using tighter geometries, transient URANS solvers to properly process vortex dynamics, and real data comparisons of wall temperatures against thermocouple measurements from the Maelstrom experimental cold-flow test. The CFD framework established here, consisting of compressible RANS modeling, conjugate heat transfer through the acrylic wall, parametric oxidizer throttling, and Roache GCI verification, provides a validated computational baseline for comparison with future experimental cold-flow testing of the Project Maelstrom engine.

References

- (1) Schmidt, Oliver, “Lecture 21-24, Finite Volume Method, SP25” MAE 185, Canvas.
https://canvas.ucsd.edu/courses/65113/files/15484478?module_item_id=2801384
- (2) Schmidt, Oliver, “Option 2: Vortex-Cooled Rocket Engine,” MAE 185, Canvas.
https://canvas.ucsd.edu/courses/65113/pages/option-2-vortex-cooled-rocket-engine?module_item_id=2702709

Appendices

Appendix A: EBU2 Computer Lab, computer specifications

Processor: Intel Core i7-12700K

RAM: 32GB

GPU: NVIDIA 970

Appendix B. Complete Simulation Results

Table B1. Mesh Independence Study Under Maximum Flow Conditions

Parameter	Coarse Mesh	Medium Mesh	Fine Mesh
Cell Count	327,970	590,087	1,644,765
Chamber Plane Temperature (K)	315.7244	315.6737	315.2896
Cross Plane Temperature (K)	302.1235	302.1238	302.1333
Nozzle Plane Temperature (K)	150.6618	146.9639	147.3924
Plenum Plane Temperature (K)	299.0291	299.0291	299.0335
Average Wall Temperature (K)	297.2634	297.2634	297.2676

Maximum Wall Temperature (K)	399.9676	399.9676	399.9675
Maximum Velocity (m/s)	666.9001	666.9001	666.9123

Table B2. Coarse Mesh Simulation Results

Parameter	Maximum Flow	Half Flow
Chamber Plane Temperature (K)	315.7244	320.0752
Cross Plane Temperature (K)	N/A	304.8850
Nozzle Plane Temperature (K)	150.6618	153.7263
Plenum Plane Temperature (K)	299.0291	300.1056
Average Wall Temperature (K)	297.2634	300.7601
Maximum Wall Temperature (K)	399.9676	399.9674
Maximum Velocity (m/s)	666.9001	645.6885

Table B3. Medium Mesh Simulation Results

Parameter	Maximum Flow	Half Flow	Quarter Flow
Chamber Plane Temperature (K)	315.6737	320.0853	332.5137
Cross Plane Temperature (K)	302.1238	304.8850	313.1158
Nozzle Plane Temperature (K)	146.9639	149.0331	224.0912

Plenum Plane Temperature (K)	299.0291	300.1056	304.5667
Average Wall Temperature (K)	297.2634	300.7601	307.8501
Maximum Wall Temperature (K)	399.9676	399.9674	399.9673
Maximum Velocity (m/s)	666.9001	645.6885	623.0048

Table B4. Fine Mesh Simulation Results

Parameter	Maximum Flow	Half Flow	Quarter Flow
Chamber Plane Temperature (K)	315.2896	320.1017	322.5137
Cross Plane Temperature (K)	302.1333	304.8402	313.1158
Nozzle Plane Temperature (K)	147.3924	149.4986	224.0912
Plenum Plane Temperature (K)	299.0335	300.0853	304.5667
Average Wall Temperature (K)	297.2676	300.7230	307.8501
Maximum Wall Temperature (K)	399.9675	399.9674	399.9673
Maximum Velocity (m/s)	666.9123	645.6885	623.0048